

YOUSSEF YASSER ELSAYED HAMED

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PROFILE

Third-year Computer Science and Artificial Intelligence undergraduate at Cairo University with a strong specialization in Deep Learning, Computer Vision, and Predictive Modeling. Holder of certifications from DeepLearning.AI, NVIDIA, and Kaggle. Experienced in building and deploying end-to-end ML solutions, delivering measurable accuracy and performance gains across diverse real-world datasets. Combines a rigorous problem-solving mindset with an entrepreneurial drive to build impactful, production-ready AI systems.

EDUCATION

Cairo University

Bachelor's in Computer Science and Artificial Intelligence

Giza, Egypt

Sep. 2023 – Sep. 2027

EXPERIENCE

AI Engineer Intern

Novus

April 2026 – Present

On-site, Giza, Egypt

- Developing and optimizing scalable deep learning pipelines, applying advanced algorithms to solve complex data-driven problems at production scale.
- Deploying AI solutions within a professional engineering environment, ensuring robust model performance, maintainable code quality, and continuous system improvements.

Machine Learning Intern

Code Alpha

April 2026

Remote

- Handwritten Character Recognition:** Built an image classification system across **47** alphanumeric classes using the EMNIST dataset using 131,000+ training images, achieving **94%** accuracy.
- Disease Prediction Model:** Developed a binary diagnostic classifier for tumor malignancy using the Breast Cancer Wisconsin dataset using 560+ clinical instances, achieving **96%** accuracy.
- Data Pipeline Optimization:** Accelerated data processing workflows using Pandas and NumPy, reducing overall processing time by **30%** while maintaining clean, normalized training data.

Competitive Programmer

Egyptian Collegiate Programming Contest (ECPC)

July 2024

Alexandria, Egypt

- Engineered optimized algorithmic solutions in C++ under strict time constraints, applying advanced data structures to tackle complex computational problems.
- Collaborated in a high-pressure team environment to prioritize problem selection, rapidly debug logical errors, and maximize competitive performance.

PROJECTS

Behavioral AI Theft Detection System [GitHub] | *Python, PyTorch, YOLOv11, SAHI, OpenCV*

April 2026

- Real-Time Vision Pipeline:** Engineered a computer vision pipeline using **YOLOv11x** for high-fidelity pose estimation and instance segmentation, actively monitoring **30** high-value target classes.
- Robust Detection & Tracking:** Integrated **SAHI** (1000×1000 tiling) to maximize small-object recall, with **ByteTrack** and a **ResNet18**-based ReID manager for persistent multi-target tracking across occlusions.
- Behavioral Classification:** Extracted **17** dynamic spatial features (grab motion, joint angles) to train a custom PyTorch **LSTM** network with an attention mechanism for classifying suspicious temporal movements.

Plant Disease Recognition System [Kaggle] | *Python, TensorFlow/Keras, CNN, OpenCV*

March 2026

- Automated Diagnosis:** Developed a deep learning system classifying **38** distinct plant disease categories using the PlantVillage dataset using 54,000+ images, achieving **99%** validation accuracy.
- Model Architecture & Training:** Designed and optimized a **Convolutional Neural Network (CNN)** pipeline for high-resolution image classification, efficiently analyzing over **54,000** images across various species.
- Comprehensive Evaluation:** Validated model performance using Confusion Matrices, Precision-Recall metrics, and per-class accuracy analysis across all 38 disease categories.

Traffic Sign Image Classifier [GitHub] | *Python, PCA, KNN, Naive Bayes, MobileNet V2*

Feb. – March 2026

- Traffic Sign Classification:** Developed a traffic sign classification system for **43** classes using 39,000+ images, achieving **98%** accuracy in road sign categories using the GTSRB dataset.

- **Dimensionality Reduction:** Applied PCA to reduce feature space while retaining **95%** of variance, improving training efficiency by **60%**.
 - **Model Comparison:** Benchmarked classical ML baselines (KNN, Naive Bayes) against a fine-tuned **MobileNet V2** CNN, boosting final accuracy to **98%**.
- Credit Risk Prediction System** [Kaggle] | *Python, Scikit-Learn, Random Forest, Pandas* Dec. 2025
- **Risk Assessment:** Developed a binary classifier to predict loan default probability using historical credit data (1,000+ records), achieving **91%** accuracy in identifying high-risk applicants.
 - **Feature Engineering:** Performed extensive preprocessing including missing value imputation, categorical encoding, and feature engineering to enhance model stability.
 - **Model Optimization:** Compared **Random Forest** and **Logistic Regression** classifiers; applied Grid Search CV for hyperparameter tuning, maximizing F1-score and ROC-AUC performance.

CERTIFICATES & AWARDS

Deep Learning Specialization <i>DeepLearning.AI (Andrew Ng) – Coursera</i>	Feb. 2026
Intermediate Machine Learning <i>Kaggle</i>	March 2026
Intro to Deep Learning <i>NVIDIA</i>	Dec. 2025
Egyptian Collegiate Programming Contest (ECPC) Participant	July 2024

STUDENT ACTIVITIES

Minders Student Activity <i>AI Workshop Participant</i>	Aug. 2025
• Participated in hands-on sessions covering the machine learning lifecycle, dataset management, and model development using Jupyter Notebooks for interactive prototyping and visualization.	

TECHNICAL SKILLS

Programming & Tools: Python, C++, SQL, Git, GitHub, Jupyter Notebook
Machine Learning & AI: Deep Learning, Computer Vision, Predictive Modeling, CNNs, Object Detection, Pose Estimation, Instance Segmentation, Random Forest, KNN, Naive Bayes, Logistic Regression
Data Science & Analysis: Exploratory Data Analysis (EDA), Feature Engineering, Data Cleaning, Data Visualization, Statistical Analysis, Model Evaluation (ROC-AUC, Precision-Recall, Confusion Matrix)
Libraries & Frameworks: PyTorch, TensorFlow/Keras, Scikit-Learn, OpenCV, YOLO (v11), MobileNet V2, ResNet, Pandas, NumPy, Matplotlib
Software Engineering: Data Structures & Algorithms, Competitive Programming, System Optimization
Soft Skills: Problem-Solving, Critical Thinking, Team Collaboration, Time Management, Adaptability, Professional Communication

LANGUAGES

Arabic: Native or bilingual proficiency
English: Professional working proficiency